

Operating System - Detailed Notes (Unit 1 to Unit 5)

UNIT 1: Operating System Overview

Definition:

- OS is an interface between user and hardware, simplifying hardware usage.

Objectives:

1. Convenience
2. Efficiency
3. Ability to evolve

Functions of OS:

- Processor, memory, file, device management, concurrency control, access control, error handling, accounting.

Evolution of OS:

1. Serial Processing
2. Batch Systems
3. Multiprogramming
4. Time-Sharing

Modern OS Characteristics:

- Multitasking, Multithreading, Virtual Memory, File & Device Management, Security, Networking, GUI/CLI, Virtualization, etc.

UNIT 2: Process and Thread Management

Process:

- A running instance of a program with unique PID, PCB, and states.

Process States:

- New, Ready, Running, Blocked, Exit (+ Suspend States)

PCB Components:

- PID, PC, CPU registers, memory limits, status, open files

Thread:

- Lightweight process; shares memory with other threads

Types:

- User-level, Kernel-level

Process vs Thread:

- Threads are lighter, faster, share resources

Modes:

- User Mode, Kernel Mode

UNIT 3: Concurrency and Deadlocks

Concurrency:

- Simultaneous process/thread execution

Problems:

- Race conditions, deadlocks, starvation, livelock

Synchronization:

- Mutual exclusion via Semaphores (wait/signal)

Problems:

- Producer-Consumer, Readers-Writers, Dining Philosophers

Deadlock:

- Caused by mutual exclusion, hold-and-wait, no preemption, circular wait

Solutions:

- Prevention, Avoidance (Banker's), Detection and Recovery

UNIT 4: Memory Management

Objectives:

- Relocation, Protection, Sharing, Logical and Physical Organization

Partitioning:

- Fixed (internal fragmentation), Dynamic (external fragmentation)

Buddy System:

- Efficient splitting and coalescing of memory

Paging:

- Fixed-size pages/frames; needs page table

Segmentation:

- Logical memory divisions

Virtual Memory:

- Disk as RAM extension

Page Replacement:

- FIFO, LRU, Optimal

UNIT 5: I/O and File Management

Part 1: I/O Management

- Categories: Human-readable, Machine-readable, Communication
- Techniques: Programmed I/O, Interrupt-driven, DMA
- Buffering: Single, Double
- Disk Scheduling: FCFS, SSTF, SCAN, C-SCAN, LOOK
- RAID: Multiple levels for redundancy & speed

Part 2: File Management

- File operations: Create, Read, Write, Delete
- Structures: Field -> Record -> File -> DB
- Organization: Pile, Sequential, Indexed, Hashed
- Directories: Maintain metadata, support search and operations
- File Sharing & Protection: ACLs, user permissions